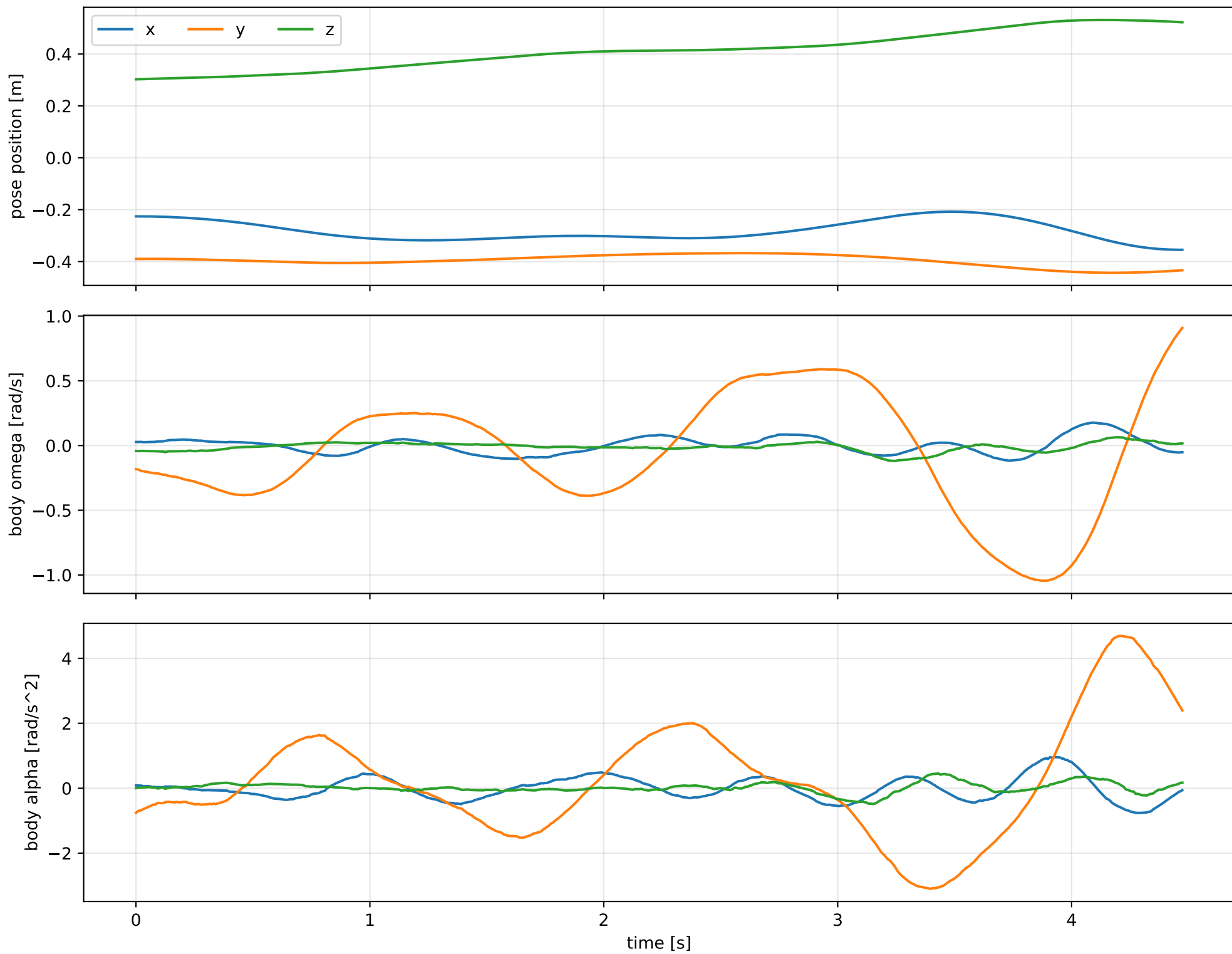
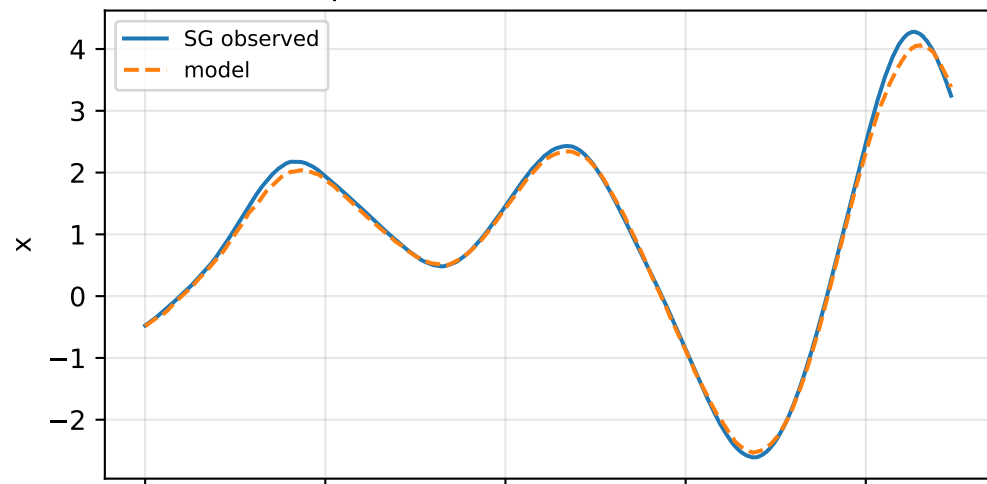
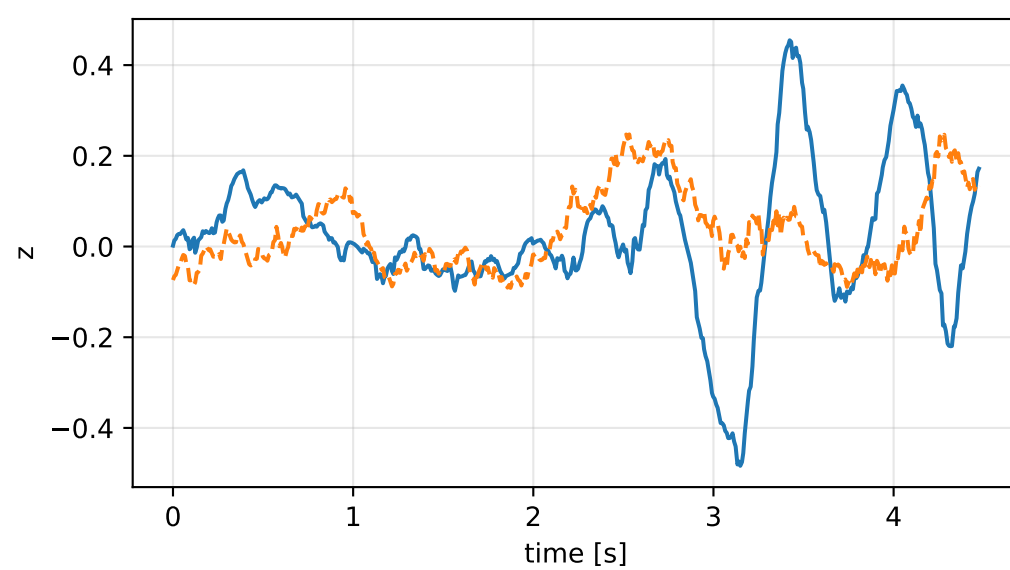
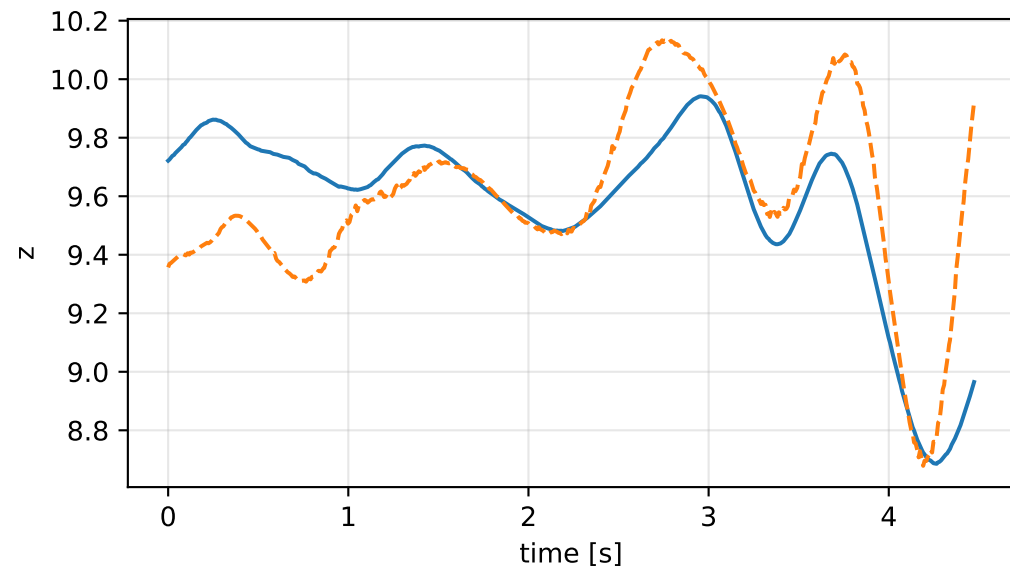
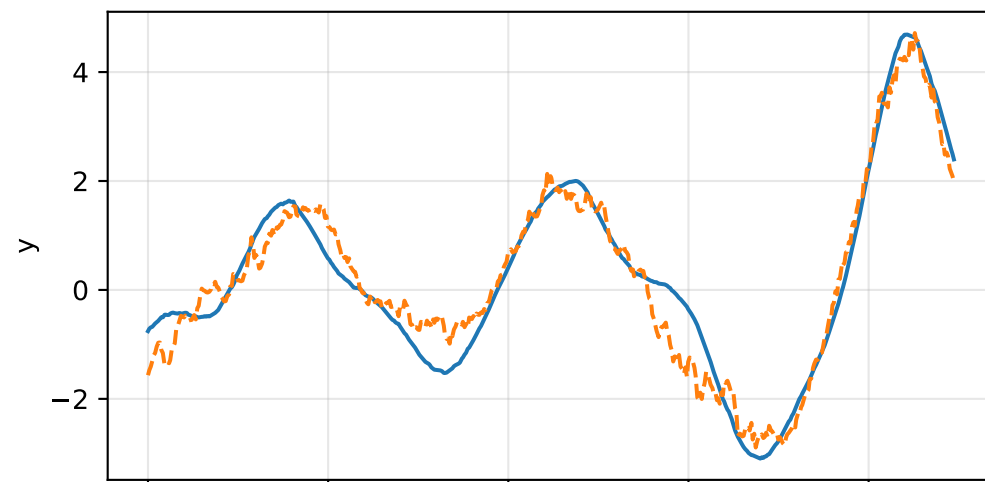
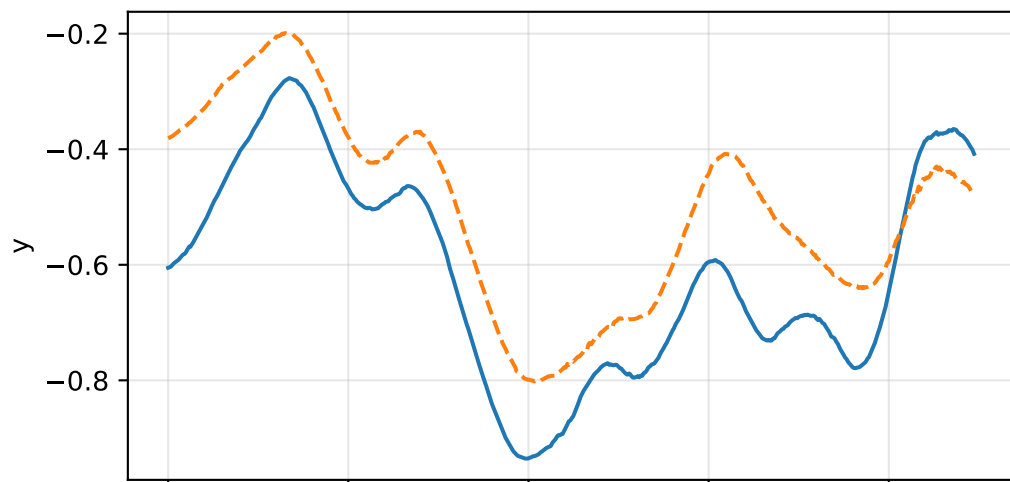
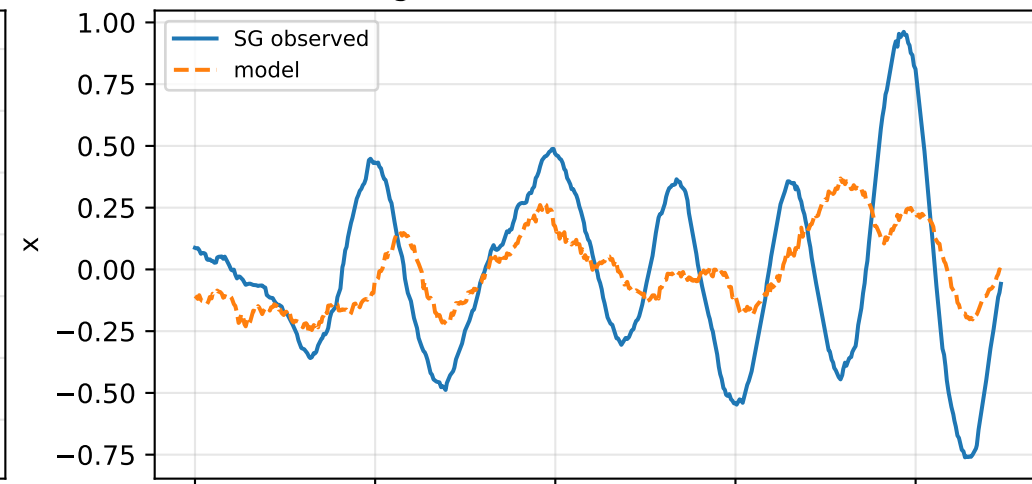


lag_pow2_depth_12: SG trajectory and derived motion

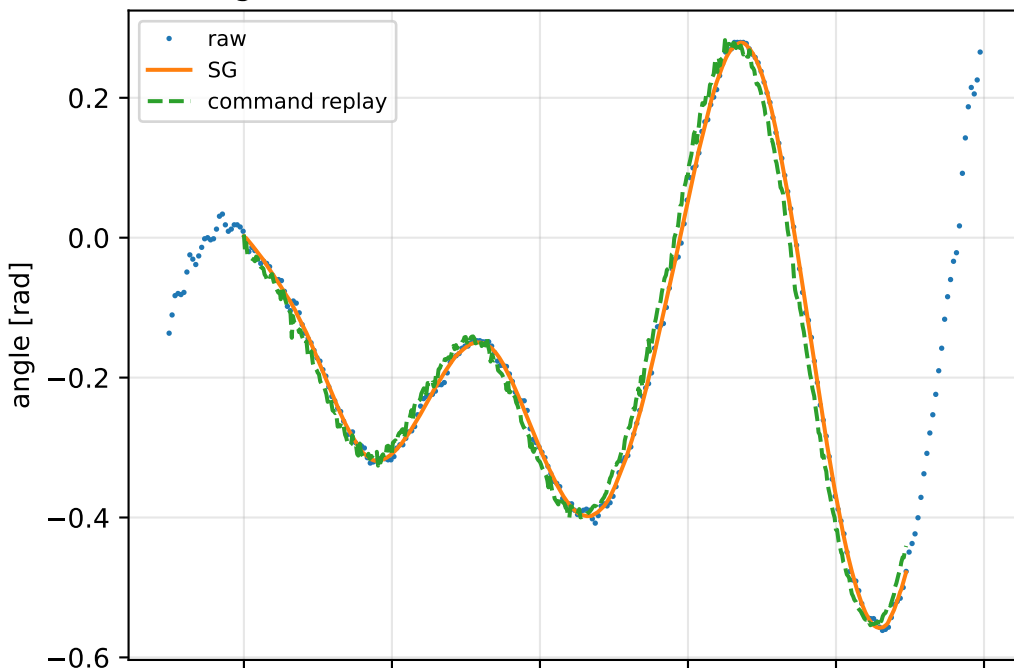


lag_pow2_depth_12: acceleration residual objective

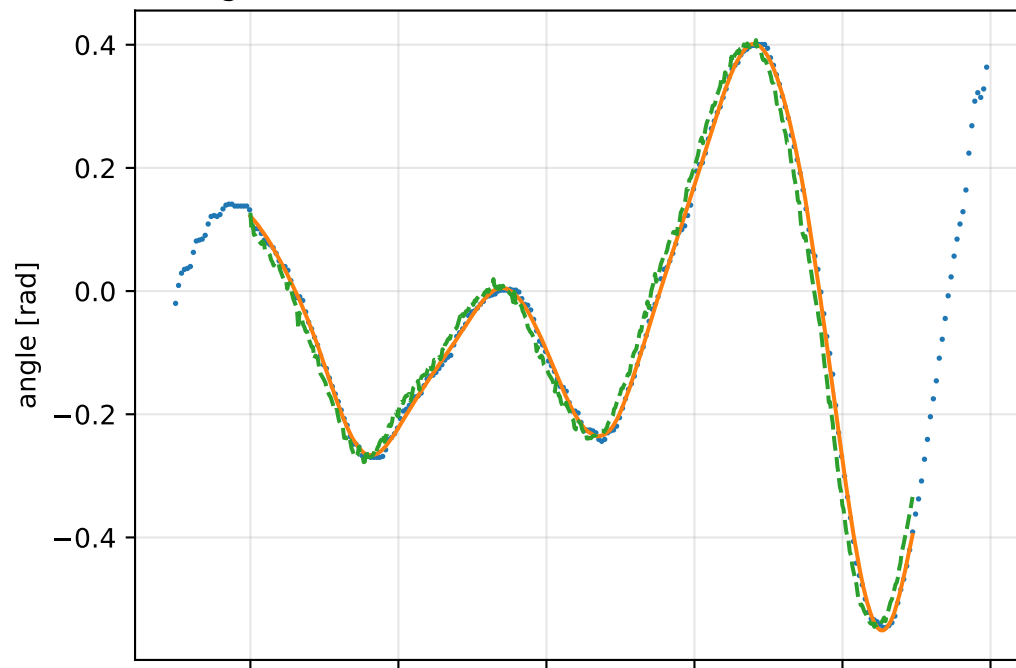
specific acceleration [m/s²]angular acceleration [rad/s²]

lag_pow2_depth_12: gimbal measurement audit (objective=measured_sg)

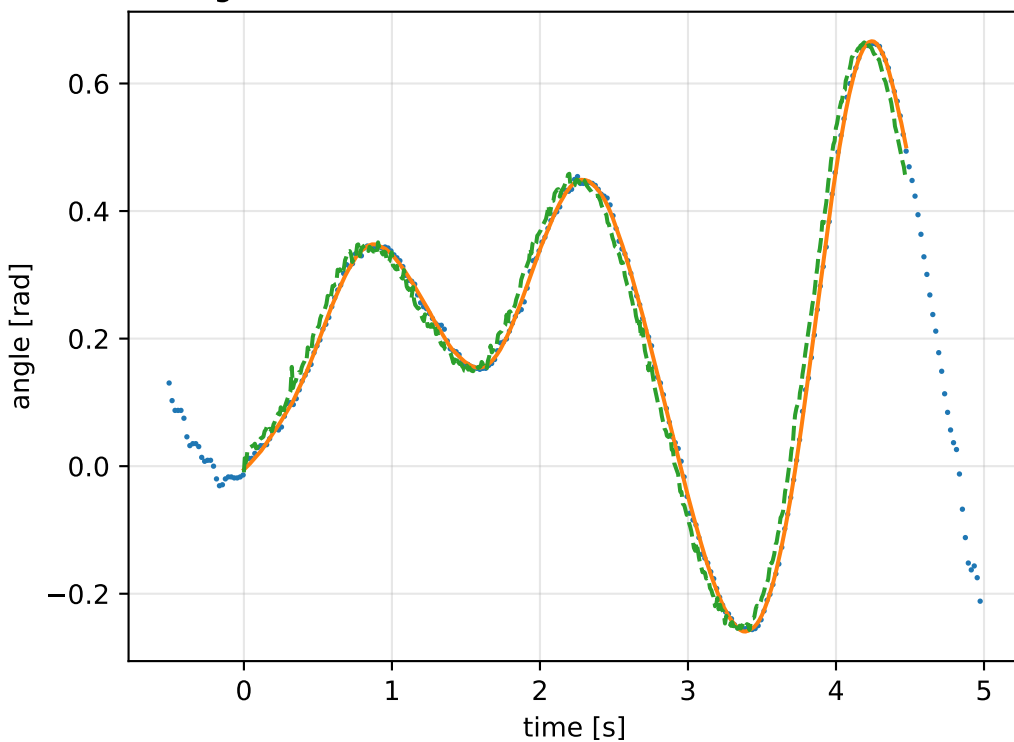
gimbal 1: RMSE=0.027, max=0.07529 rad



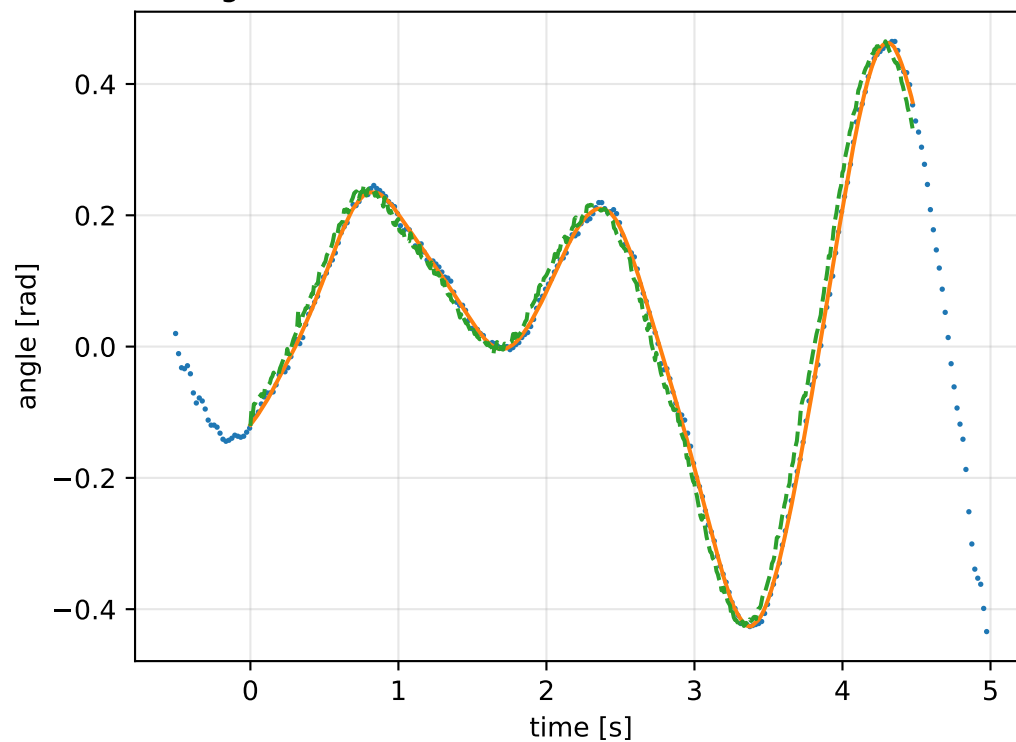
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

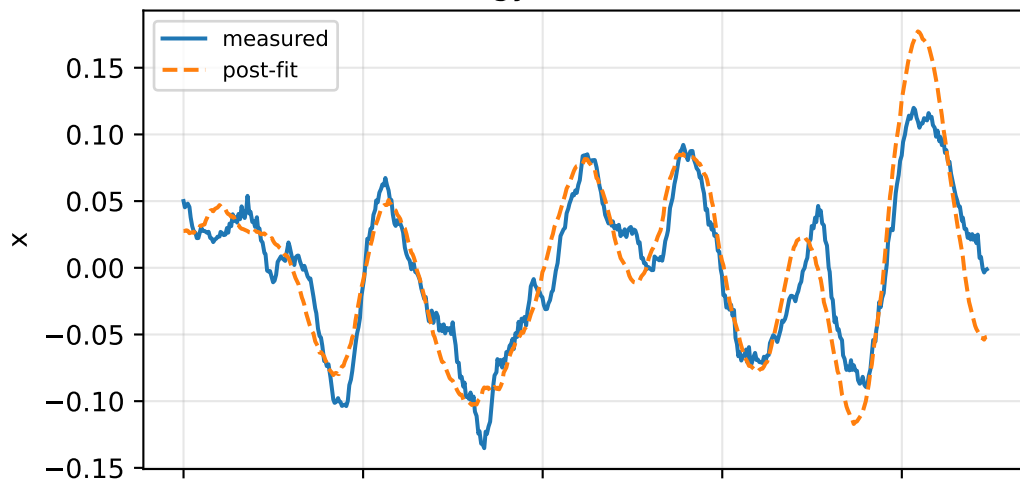
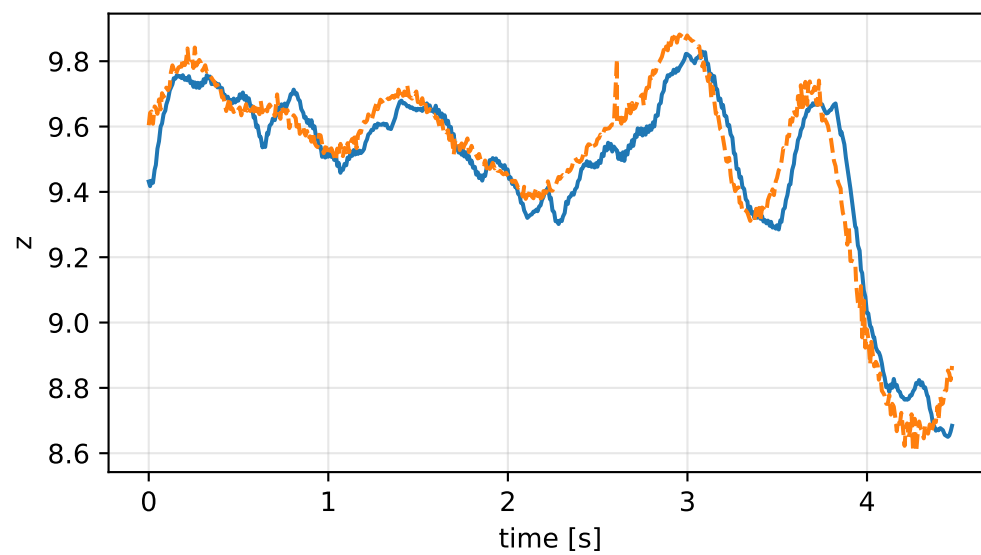
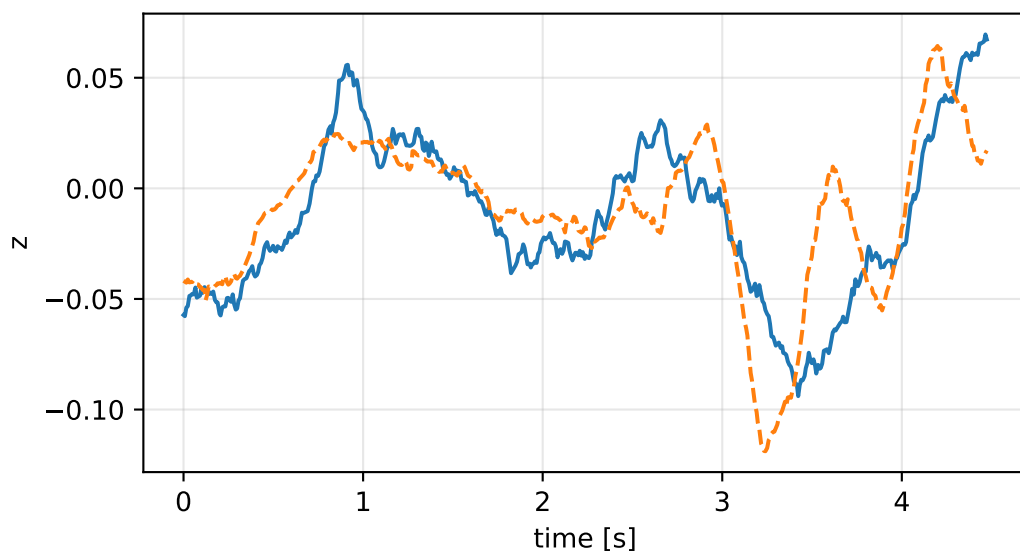
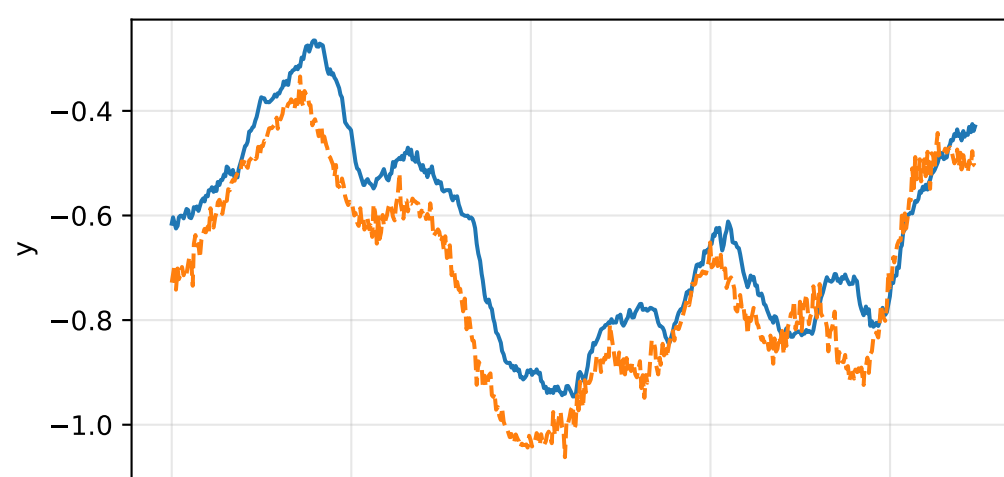
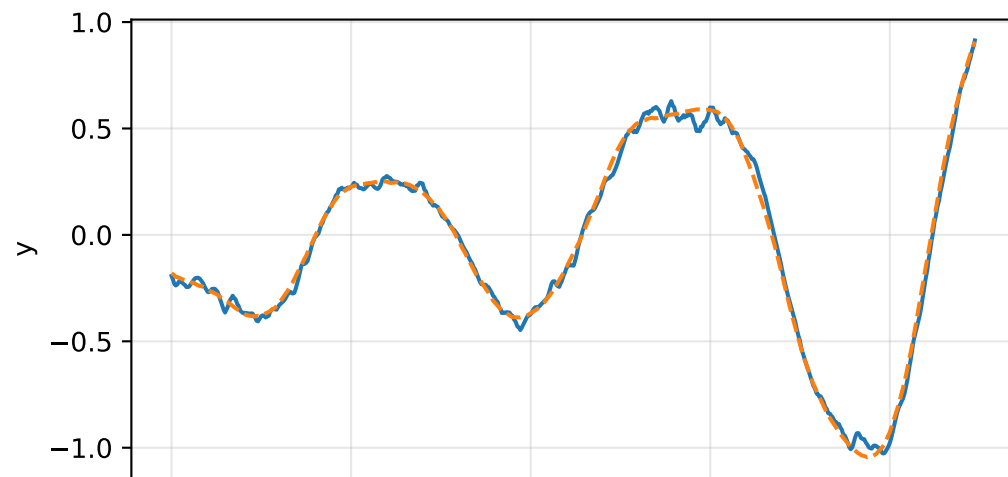
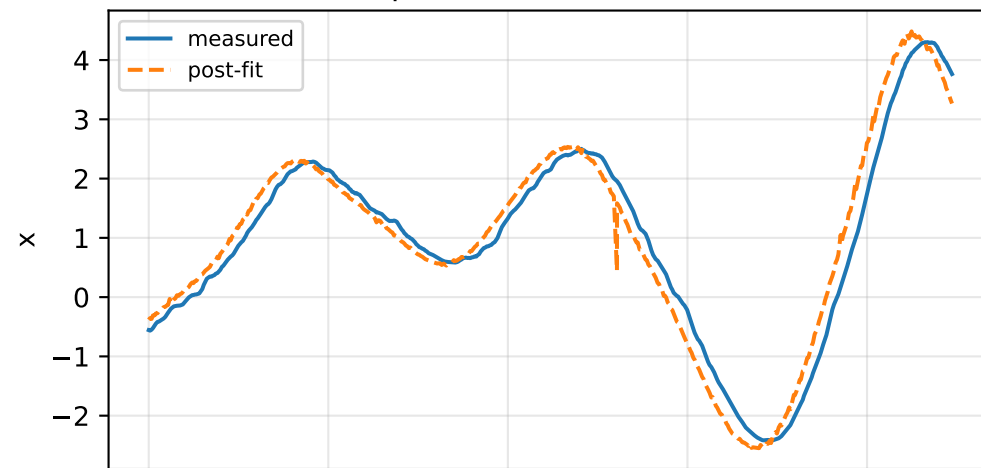


gimbal 4: RMSE=0.02598, max=0.07261 rad

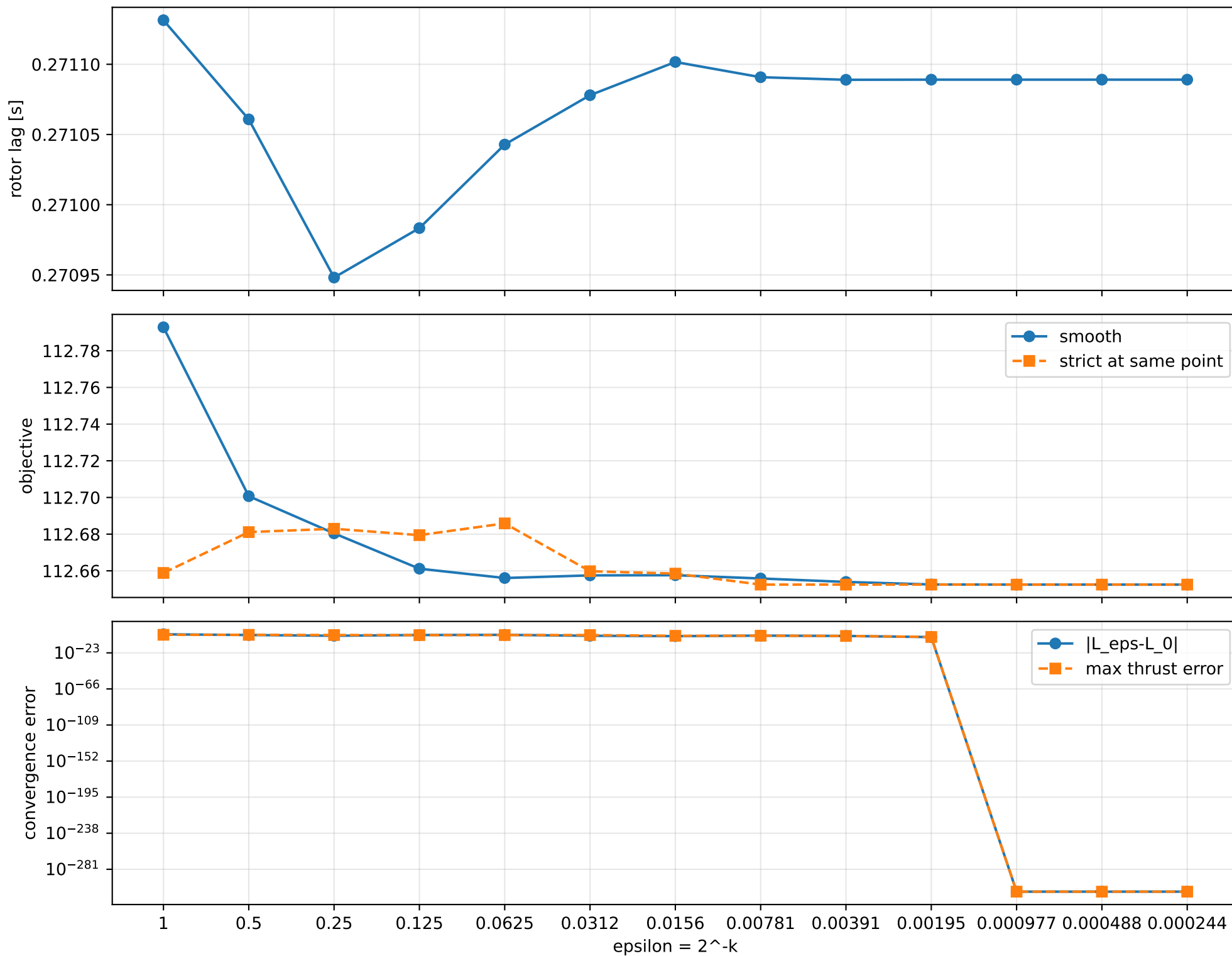


lag_pow2_depth_12: IMU comparison (diagnostic only)

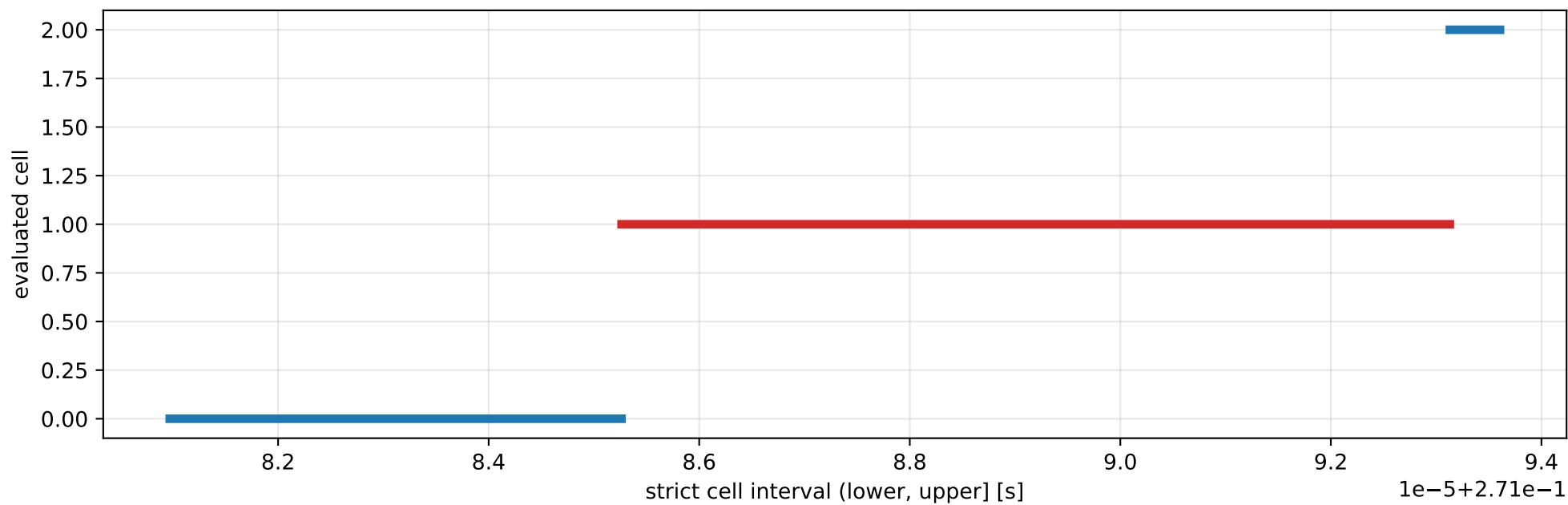
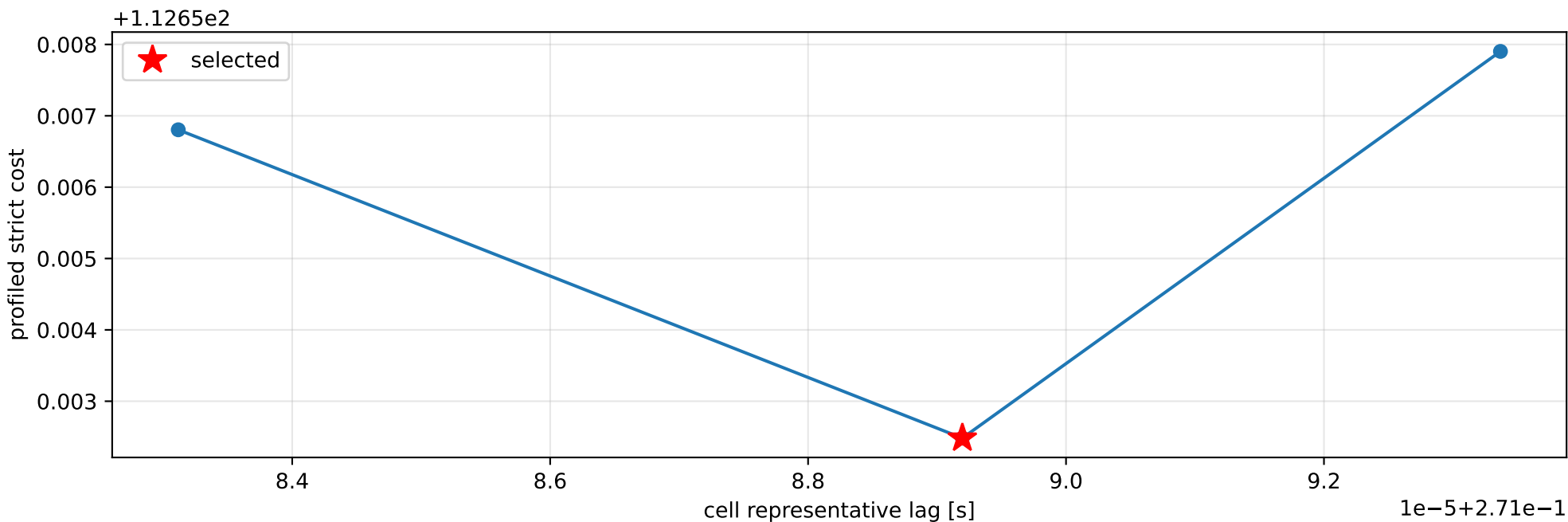
gyro [rad/s]

specific force [m/s²]

lag_pow2_depth_12: smooth-to-strict continuation



lag_pow2_depth_12: exact strict-ZOH lag cells



selected (0.271085262, 0.27109313] s; final smooth 0.271089061 s; data boundary=False

lag_pow2_depth_12: parameter estimate

Case: lag_pow2_depth_12

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 7.5204299

inertia [kg m²]:

```
[[ 0.55443743  0.01372526 -0.00323578]
 [ 0.01372526  0.2616835  -0.08172249]
 [-0.00323578 -0.08172249  0.79125054]]
```

CoG body [m]: [-0.02643501 0.01133914 0.00941532]

force effectiveness: [2.87131601 2.32387849 2.56341348 3.07927321]

scale-free J/m [m²]:

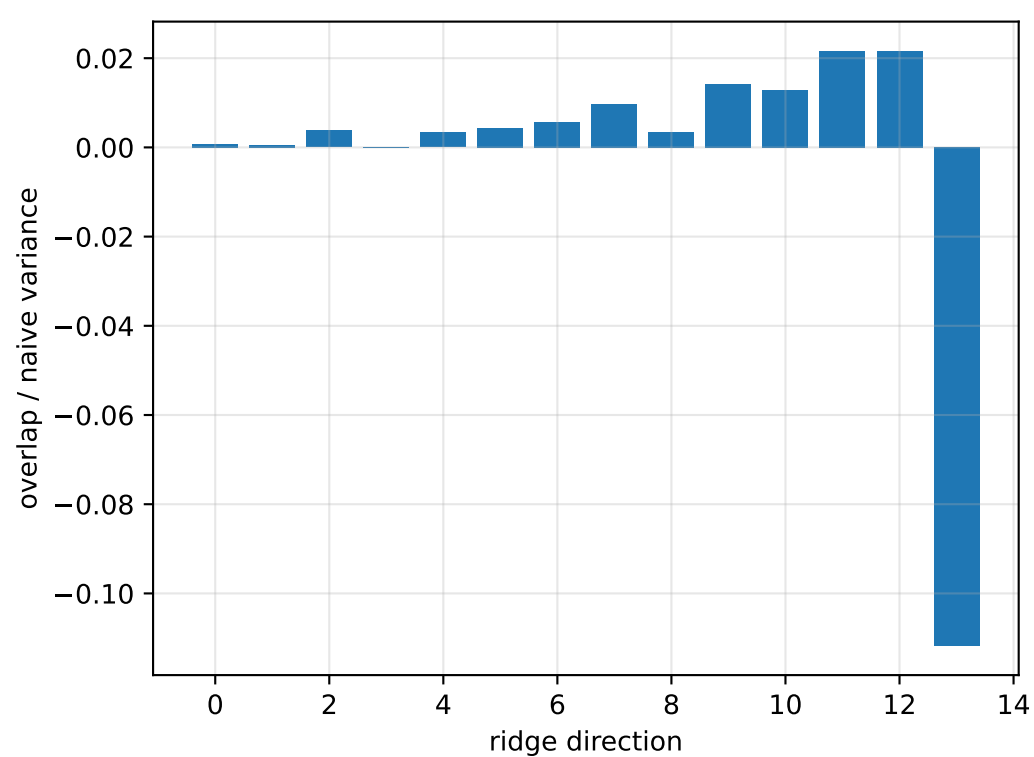
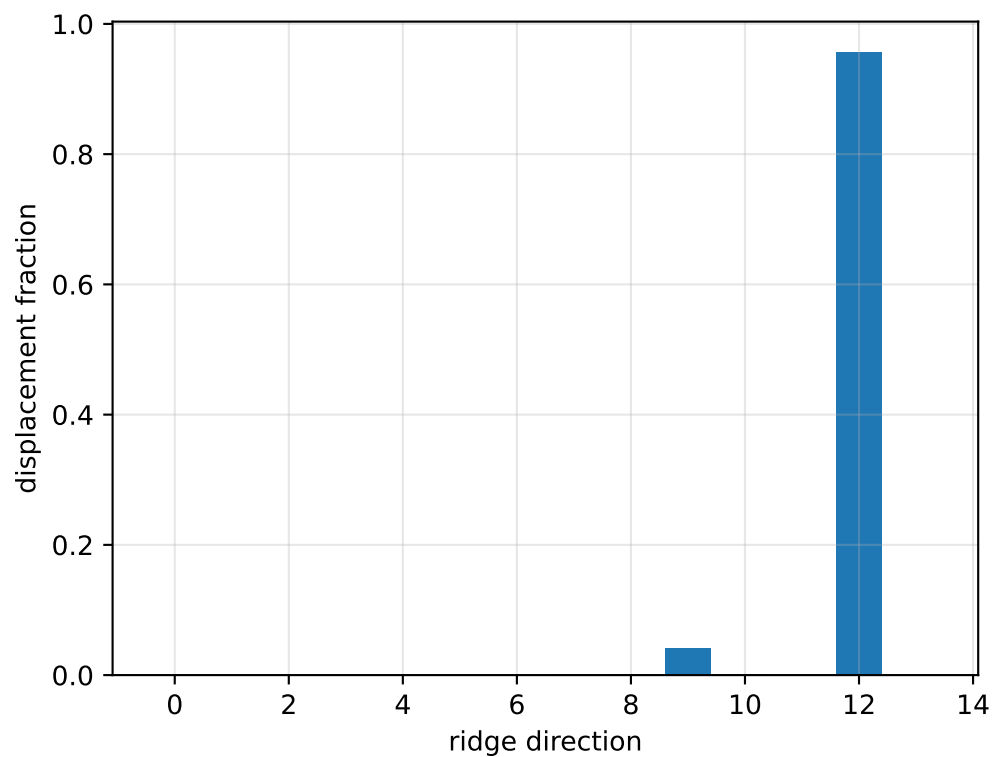
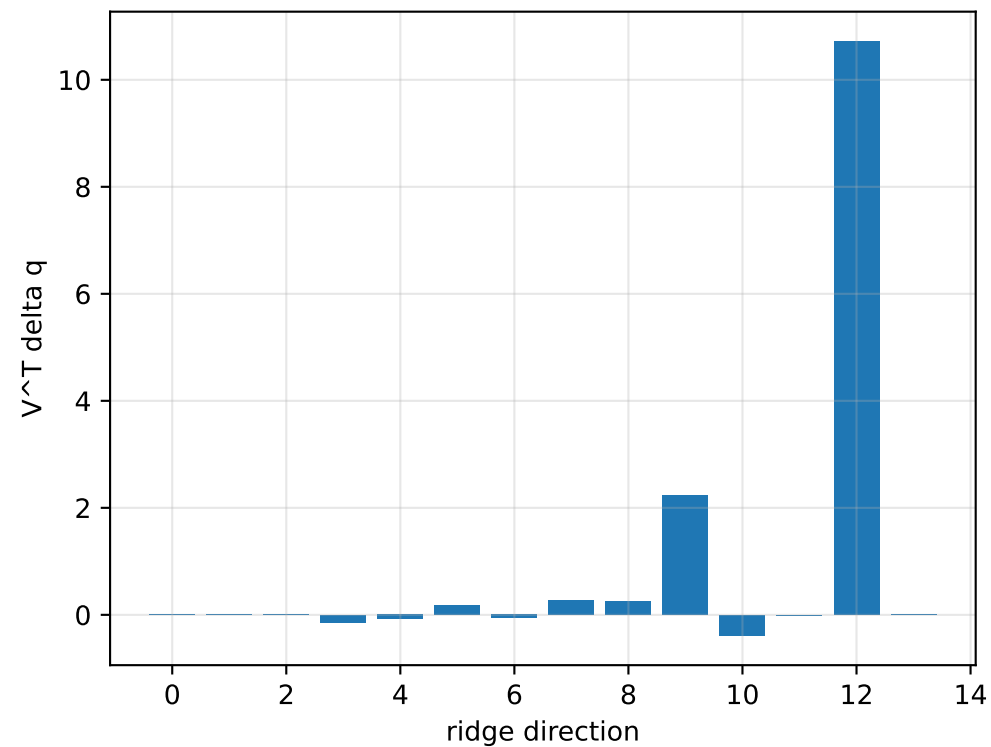
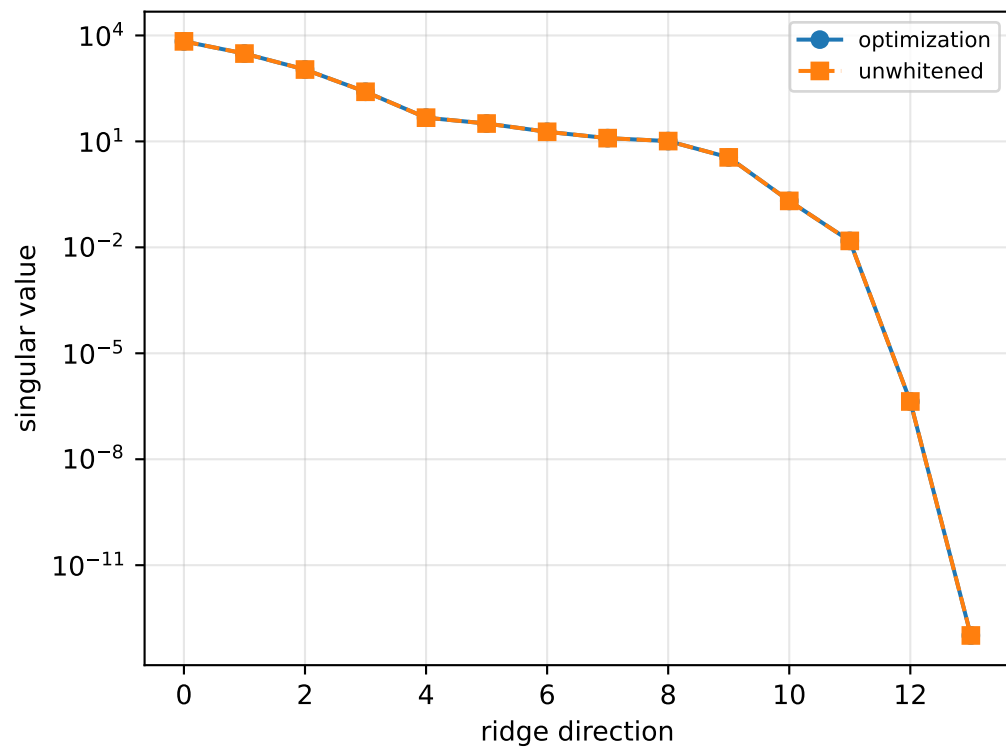
```
[[ 0.07372417  0.00182506 -0.00043027]
 [ 0.00182506  0.03479635 -0.01086673]
 [-0.00043027 -0.01086673  0.10521347]]
```

scale-free f/m: [0.38180211 0.30900873 0.34085997 0.40945441]

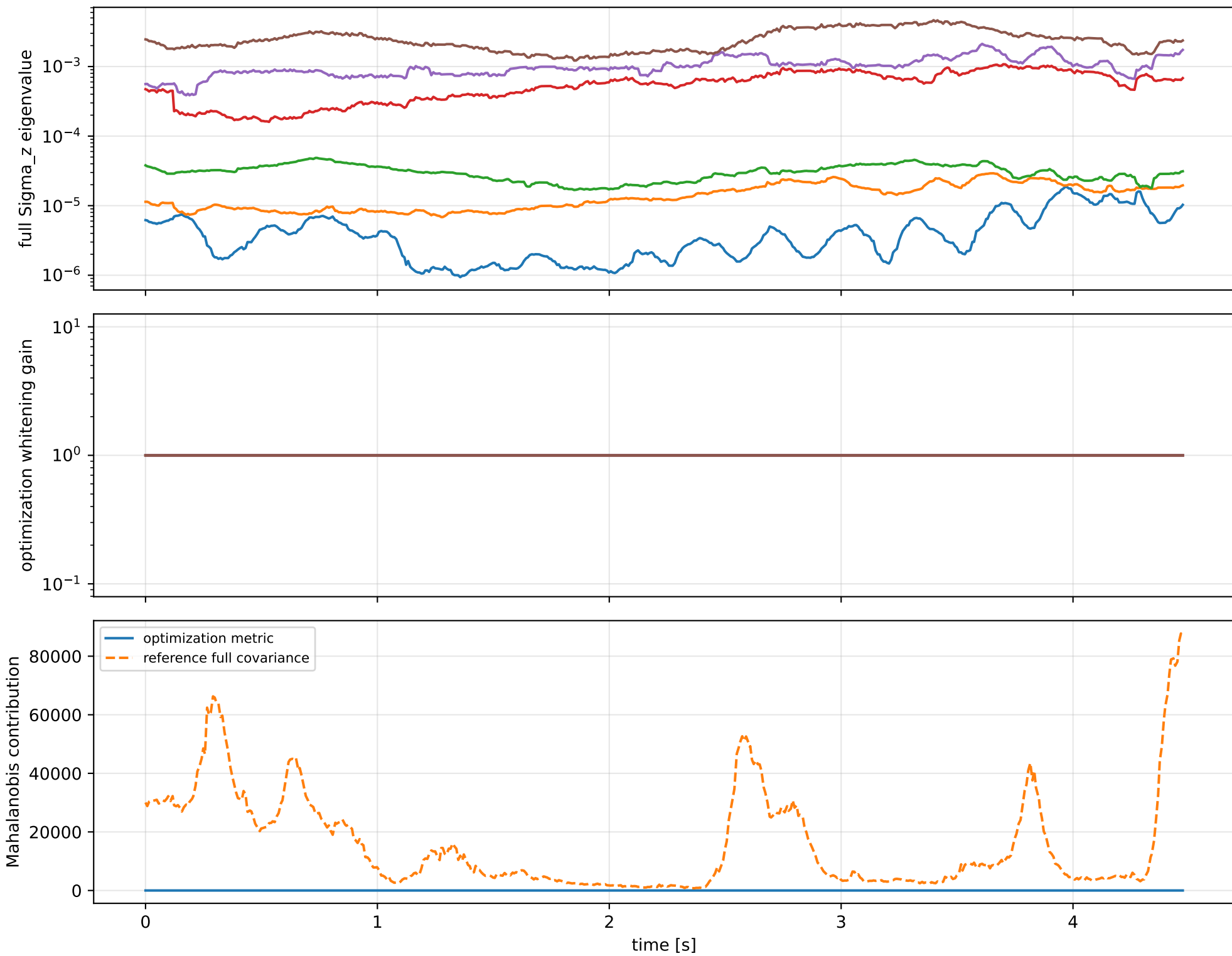
rotor lag cell: (0.271085262, 0.27109313) s

representative: 0.271089196 s

lag_pow2_depth_12: ridge spectrum (rank 13, nullity 1, $||\mathbf{J}\mathbf{v}||=1.56\text{e-}13$)



lag_pow2_depth_12: optimization vs reference covariance



lag_pow2_depth_12: wrench / closure (force max $2.3\text{e-}14$, torque max $5.41\text{e-}16$)

